

Recitation: Factoring, Roots, & Discriminants Solutions

Problem 1. If any of the following polynomials factor, factor them completely:

(a) $x^2 - 10x - 24$

(f) $x^2 + 1$

(b) $25x^3 - 4x$

(g) $x^4 + x^3 - 2x^2$

(c) $x^2 - 7x - 6$

(h) $3x^2 - x - 2$

(d) $x^3 + 8$

(i) $x^3 - 2x^2 + 5x - 10$

(e) $4x^2 + 8x + 3$

(j) $32x^2 - 76x + 35$ [Hint: This has roots $\frac{5}{8}, \frac{7}{4}$.]

Solutions.

(a) Because $-12 \cdot 2 = -24$ and $-12 + 2 = -10$, we have...

$$x^2 - 10x - 24 = (x - 12)(x + 2)$$

(b) Factoring out x , we have $25x^3 - 4x = x(25x^2 - 4)$. But the second term we recognize as a difference of perfect squares, $a^2 - b^2 = (a - b)(a + b)$, with $a = 5x$ and $b = 2$. So, we have...

$$25x^3 - 4x = x(25x^2 - 4) = x(5x - 2)(5x + 2)$$

(c) This does not factor (nicely). Observe the discriminant $D = b^2 - 4ac = (-7)^2 - 4(1)(-6) = 73$ is not a perfect square.

(d) This is a sum of perfect cubes: $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$, with $a = x$ and $b = 2$, so we have...

$$x^3 + 8 = (x + 2)(x^2 - 2x + 4)$$

(e) Observe $4 \cdot 3 = 12$ and then observe $2 \cdot 6 = 12$ and $2 + 6 = 8$. So, we have...

$$4x^2 + 8x + 3 = (4x^2 + 2x) + (6x + 3) = 2x(2x + 1) + 3(2x + 1) = (2x + 1)(2x + 3)$$

(f) This is a sum of perfect squares—which *never* factor over $\mathbb{R}$. The discriminant is also $D = b^2 - 4ac = 0^2 - 4(1)(1) = -4$, which indicates it does not factor. Over the complex numbers, it factors as $(x - i)(x + i)$.

(g) Factoring out the common x^2 , we have $x^4 + x^3 - 2x^2 = x^2(x^2 + x - 2)$. As $-1 \cdot 2 = -2$ and $-1 + 2 = 1$, we have $x^2 + x - 2 = (x - 1)(x + 2)$. So, we have...

$$x^4 + x^3 - 2x^2 = x^2(x^2 + x - 2) = x^2(x - 1)(x + 2)$$

(h) Observe $3 \cdot -2 = -6$, and observe $-3 \cdot 2 = -6$ and $-3 + 2 = -1$. So, we have...

$$3x^2 - x - 2 = (3x^2 - 3x) + (2x - 2) = 3x(x - 1) + 2(x - 1) = (x - 1)(3x + 2)$$

(i) Because of the four terms, we suspect this may factor by grouping (which we use when $a \neq 1$). Observe...

$$x^3 - 2x^2 + 5x - 10 = (x^3 - 2x^2) + (5x - 10) = x^2(x - 2) + 5(x - 2) = (x - 2)(x^2 + 5)$$

(j) Using the Fundamental Theorem of Algebra, we know a quadratic factors as $a(x - r_1)(x - r_2)$, where r_1, r_2 are the roots. So, we have...

$$32x^2 - 76x + 35 = 32 \left(x - \frac{5}{8}\right) \left(x - \frac{7}{4}\right) = 8 \left(x - \frac{5}{8}\right) \cdot 4 \left(x - \frac{7}{4}\right) = (8x - 5)(4x - 7)$$

Problem 2. If $p(x)$ is a polynomial has roots $x = -1, -1, -1, 1, 3$ and y -intercept 5, determine the polynomial. Also, determine the domain and range of $p(x)$.

By the Fundamental Theorem of Algebra, we know that a polynomial factors as $a(x-r_1)(x-r_2)\cdots(x-r_{n-1})(x-r_n)$, where the r_i are the roots and a is the leading coefficient. So, using the given information, we have...

$$p(x) = a(x - (-1))(x - (-1))(x - (-1))(x - 1)(x - 3) = a(x + 1)^3(x - 1)(x - 3)$$

But we know the y -intercept is $y = 5$, i.e. $y = 5$ when $x = 0$. But then...

$$p(x) = a(x + 1)^3(x - 1)(x - 3)$$

$$p(0) = a(0 + 1)^3(0 - 1)(0 - 3)$$

$$5 = 3a$$

$$a = \frac{5}{3}$$

Therefore, $p(x) = \frac{5}{3}(x + 1)^3(x - 1)(x - 3)$.

Problem 3. Find all possible solutions to the following:

$$\frac{5}{2-x} = x$$

$$\frac{5}{2-x} = x$$

$$5 = x(2-x)$$

$$5 = 2x - x^2$$

$$x^2 - 2x + 5 = 0$$

Using the Quadratic Formula with $a = 1, b = -2, c = 5$, we have...

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-(-2) \pm \sqrt{(-2)^2 - 4(1)5}}{2(1)} = \frac{2 \pm \sqrt{-16}}{2} = \frac{2 \pm i\sqrt{16}}{2} = \frac{2 \pm 4i}{2} = 1 \pm 2i$$

Problem 4. Explain why the polynomial $100x^2 - 400x + 100$ does not factor 'nicely'. Then use the quadratic formula to factor the polynomial completely.

Observe $100x^2 - 400x + 100$ factors 'nicely' if and only if $x^2 - 4x + 1$ factors 'nicely' because $100x^2 - 400x + 100 = 100(x^2 - 4x + 1)$. The polynomial $x^2 - 4x + 1$ has discriminant $D = b^2 - 4ac = (-4)^2 - 4(1)(1) = 12$, which is not a perfect square. So, $x^2 - 4x + 1$ does not factor 'nicely.' Now, using the quadratic formula, we know that $x^2 - 4x + 1$ has roots...

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-(-4) \pm \sqrt{(-4)^2 - 4(1)1}}{2(1)} = \frac{4 \pm \sqrt{12}}{2} = \frac{4 \pm 2\sqrt{3}}{2} = 2 \pm \sqrt{3}$$

Then, using the Fundamental Theorem of Algebra, we have...

$$100x^2 - 400x + 100 = 100(x - (2 + \sqrt{3}))(x - (2 - \sqrt{3}))$$